WASP-52b

R-1153

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_201103` · created 2026-07-31T01:27:19+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459156.7593 +/- 0.0058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.211 +/- 0.062
Transit depth (Rp/Rs)^2	4.46 +/- 2.63 [%]
Orbital Inclination (inc)	82.9 +/- 2.73
Airmass coefficient 1 (a1)	0.8 +/- 0.47
Airmass coefficient 2 (a2)	0.24 +/- 0.44
Scatter in the residuals of the lightcurve fit is	55.36 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	3.5
Transit Duration (day)	0.049 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.211 ± 0.062 vs 0.1646 (deviation 0.55σ)
Duration fit vs published	0.049 d vs 0.0758 d (deviation 0.73σ)
Mid-time O–C	-2.5 min
Depth SNR	2.5
Residual scatter	55.36 %

Light curve

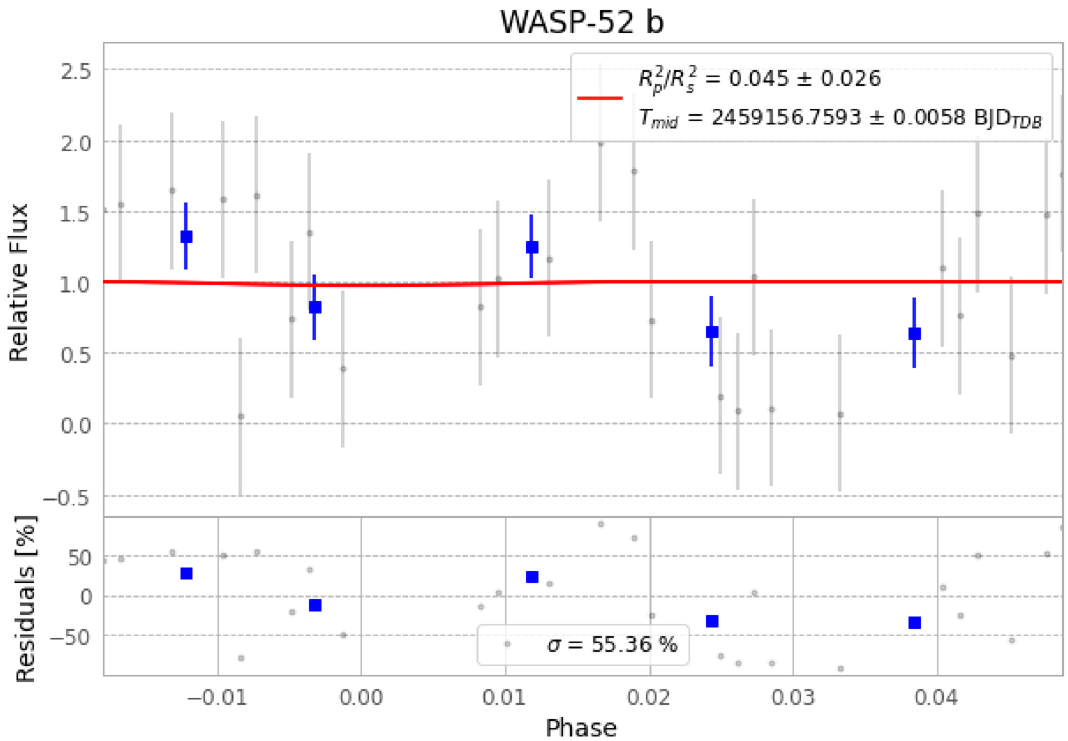

Figure 1 — FinalLightCurve_WASP-52 b_2020-11-02.png (SHA-256 9a4749485ce7eae0…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [607, 87], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 381258345d161d50…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

57 FITS frames (37 MB), `WASP-52201103052112.FITS` → `WASP-52201103080912.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-201103.log (eb655336d4f3…), FinalLightCurve_WASP-52 b_2020-11-02.png (9a4749485ce7…), FinalLightCurve_WASP-52 b_2020-11-02.csv (e00f6def44d8…)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-31 01:27Z.